Brain Tumor Report

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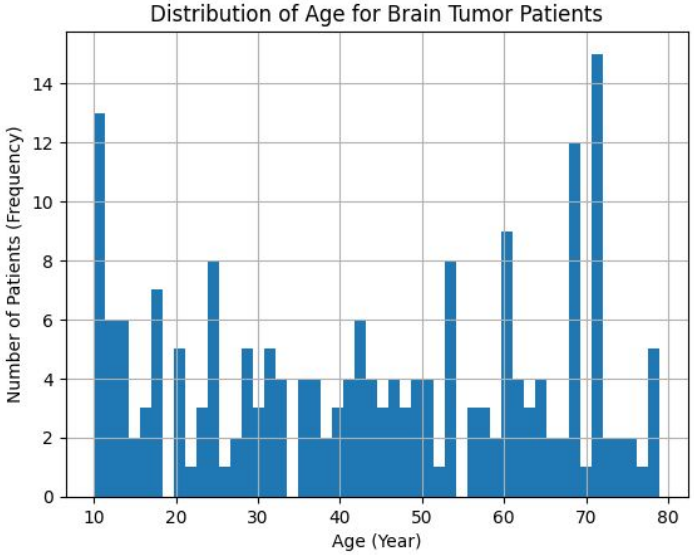
Data Background

- Three Datasets (.csv file) from Kaggle
  - Patient Information
  - Treatment Outcome
  - Tumor Information
- All cleaned and ready to be analyzed
  - No missing value
  - Mostly categorical values
- 200 values each, connected with other by patient ID
- Change categorical values into ordinal value
  - Binary categorical (Yes/No) → Change to 0/1
  - Categorical with more than 2 categories signify a spectrum → Change to spectrum of number (i.e; “Outcome_Status" has 3 categories: Recovered (map to 1), Stable (map to 0) and Critical (map to -1)

Patient Information Dataset

- Description: Include patient ID, age, and gender
- Two age group that appear the most
  - Young children and teenager age 10-12
  - Elderly of age 65-72
- Upon examining the age and gender, there is no significant difference between the distribution of age between the male and female gender

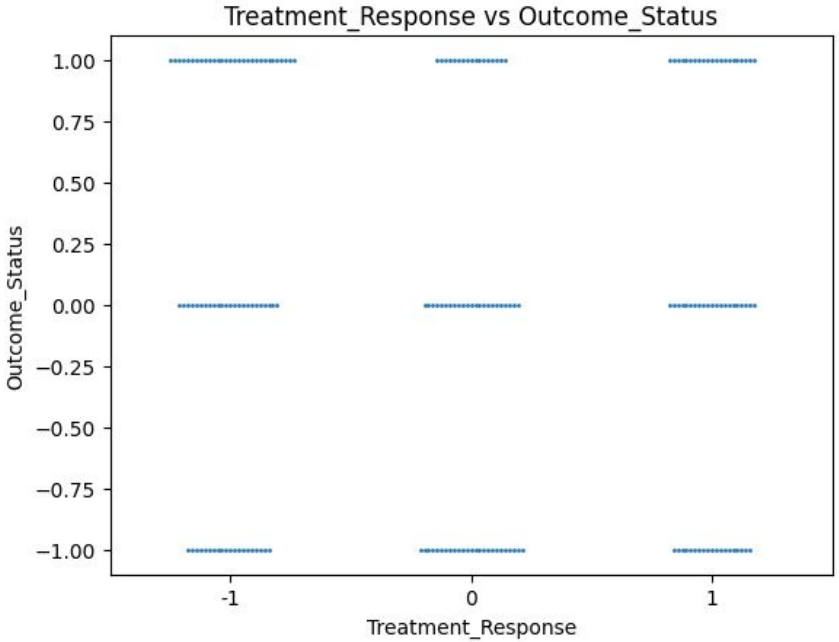


Treatment Outcome Dataset

- Description: Information about the type of treatments each patient get along with their treatment responses, survival months, and outcomes.
- Questions to Consider
  - Are there any correlation between treatment response and outcome status?
  - Does tumor progression associate with survival month/treatment response/outcome status?
  - Does recurrence associate survival month/treatment response/outcome status?
  - Does early detection help with survival month/treatment response/outcome status?

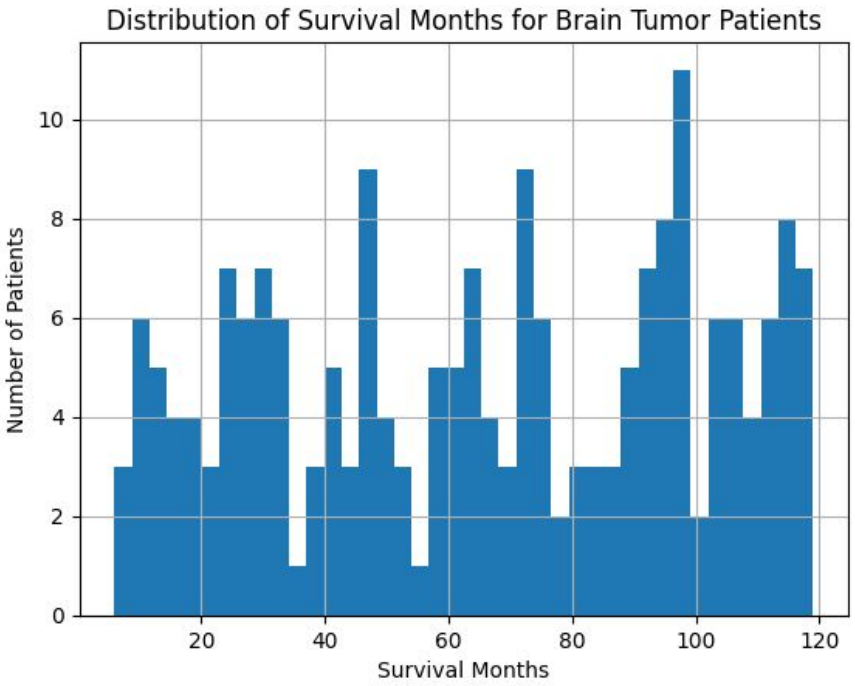
Correlation Between Treatment Response and Outcome Status

- Treatment response
  - 1: Good
  - 0: Moderate
  - -1: Poor
- Outcome status
  - 1: Recovered
  - 0: Stable
  - -1: Critical
- r = -0.0582
- There is little to no relationship



Assumption: Survival Months

- If patients are still alive after the data collection phase ended, then there will be potential data fraud, since the survival months will not be correct data anymore
- Looking at the distribution of survival months, there is no outliner at the maximum, so it's reasonable to assume that there are no such data fraud



Tumor Progression vs Survival Month/ Treatment Response/Outcome Status?

- Tumor Progression vs Survival Month
  - Independent 2-sample t-test
  - p-value = 0.554
  - No significant difference between distributions of survival month for patient with tumor progression or patient without tumor progression
- Tumor Progression vs Treatment Response and Outcome Status
  - Chi-Square Test for Homogeneity
  - p-value
    - Tumor Progression vs Treatment Response: p = 0.980
    - Tumor Progression vs Outcome Status: p = 0.596
  - No significant difference between distributions of treatment response and outcome status for patient with tumor progression or patient without tumor progression.

Recurrence vs Survival Month/ Treatment Response/Outcome Status?

- Recurrence vs Survival Month
  - Independent 2-sample t-test
  - p-value = 0.998
  - No significant difference between distributions of survival month for patient with recurrence or patient without recurrence
- Recurrence vs Treatment Response and Outcome Status
  - Chi-Square Test for Homogeneity
  - p-value
    - Recurrence vs Treatment Response: p = 0.505
    - Recurrence vs Outcome Status: p = 0.086
  - No significant difference between distributions of treatment response and outcome status for patient with recurrence or patient without recurrence.

Early Detection vs Survival Month/ Treatment Response/Outcome Status?

- Early Detection vs Survival Month
  - Independent 2-sample t-test
  - p-value = 0.296
  - No significant difference between distributions of survival month for patient with early detection or patient without early detection
- Early Detection vs Treatment Response and Outcome Status
  - Chi-Square Test for Homogeneity
  - p-value
    - Recurrence vs Treatment Response: p = 0.137
    - Recurrence vs Outcome Status: p = 0.375
  - No significant difference between distributions of treatment response and outcome status for patient with early detection or patient without early detection

Tumor Information Dataset

- Description: Information about the tumor information (type, grade, size and location), the symptoms (headache, seizure, nausea, etc.), and the risk factors of each patient
- Question to consider
  - Does the tumor type has any effect on the tumor size?
  - Correlation between tumor grade and tumor size

Tumor Type vs Tumor Size

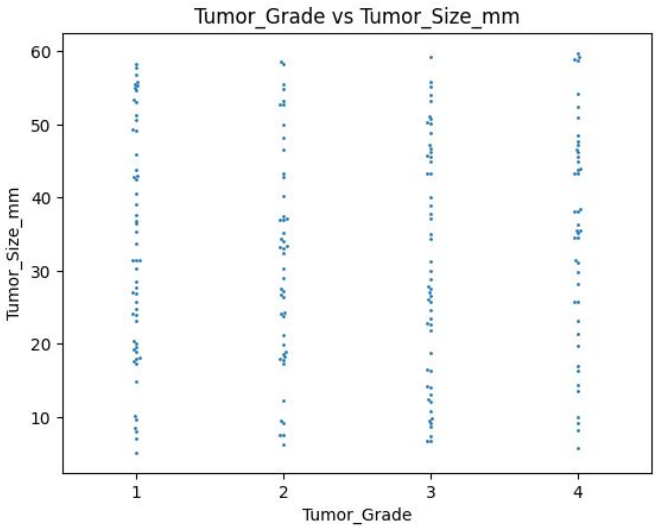
- Categorical (>2 categories) and Numerical Variable
- One-way ANOVA Test
- Conditions
  - Independence
  - Normality
    - Each group has at least 57 data → Central Limit Theorem
  - Homogeneity of variance:
    - Levene's Test:
      - H0: Equal variances across groups
      - Ha: At least two groups don't have equal variances
    - p-value = 0.904
    - Conclusion: Unable to reject the null hypothesis of equal variance

Tumor Type vs Tumor Size

- One-way ANOVA test
- p-value = 0.759
- Conclusion: There are no significant difference between the distribution of tumor size across groups of patients who have tumor type Glioma, Meningioma, and Pituitary

Tumor Grade vs Tumor Size

- Tumor Grade: Grade I, II, III, and IV
- Change Tumor Grade to Ordinal Value
  - I → 1
  - II → 2
  - III → 3
  - IV → 4
- r = 0.0327
- There is little to no relationship between tumor grade and tumor size



Merge Dataset

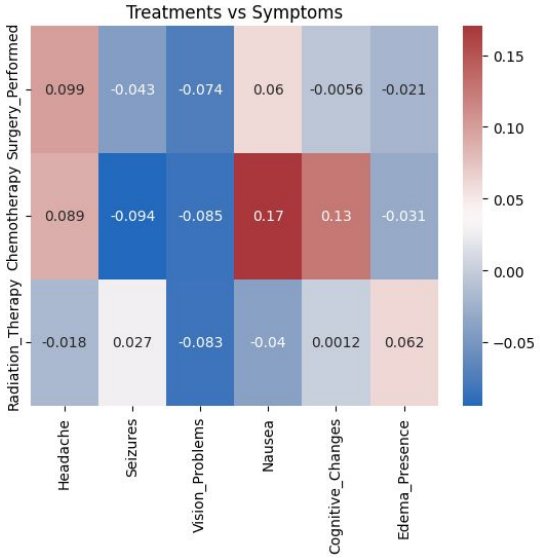
- Merge/Join all three datasets
- Use Patient ID as a key to merge dataset together
- Goal: Find correlation between
  - Age and survival month
  - Treatments and symptoms
  - Family history and generic risk

Age vs Survival Month

- Age distribution is bi-model
- According to National Brain Tumor Society, 81.7% of all primary brain tumors occurs in the adult (40+) population
- Split age into 2 groups
  - Less than or equal to 40
  - More than 40
- Perform independent 2-sample t-test
- p-value: 0.349
- Conclusion: There is no significant difference in the distribution of survival month between patients whose age is 40 or less or more than 40.

Treatments vs Symptoms

- Both are categorical variables
- Each patient can have multiple treatments and symptoms
  → One-hot encoding
- Use heatmap to represent correlation
- There is little to no correlation
- Chemotherapy usually associate with nausea, headache, and cognitive change more than other treatments



Family History vs. Generic Risk

- Scatter plot and correlation
  - r = -0.020
  - Very weak and negative relationship
- Chi-Square test
  - p-value = 0.190
  - Conclusion: There are no significant differences between the distribution of generic risk between brain tumor patients with and without family history of brain tumor